The a'_{hdm} mirror ($\kappa\approx 0$): three parallel tracks

The whole derivation answers one question: *if the variance-ratio reading $\hat{a}'_{hdm}$ is fed to the a' update, does the error $e_{a'_h} = a'_{hd} - \hat{a}'_{hd}$ shrink (converge) or drift?* The answer is built from three parallel tracks – **True** (how a_h, a', a_r really evolve), **Estimate** (how $\hat{a}'_{hd}, \hat{a}_r, \hat{a}'_{hdm}$ are computed online), and **Error** = True − Estimate. Section 3 is the bridge: it shows the Estimate track is a *mirror* of the controller's own $\hat{a}'_{hd}$, not a measurement of a'_{hd}.

0. Definitions and $[k]$ timing

$$h[k] = h_d[k] - \delta h[k], \qquad \Delta h_d[k] = h_d[k+1] - h_d[k]$$

$$a'_{hd}[k] := \left.\frac{da_h}{dh}\right|_{h_d[k]}, \qquad a_r[k] := a_h[k] - a_h\big(h_d[k]\big)$$

$$\delta h[k+1] = \lambda_c\, \delta h[k] - F_{dh}[k]\, e_{a_h}[k] - \varepsilon_h[k]$$

$$a_{var} = a_{cov} = 0.05, \qquad T_s = 1/1600, \qquad \beta = 1 - T_s/\tau,\ \tau = 0.35\,\mathrm{s}\ \Rightarrow\ \beta = 0.99821$$

$[k]$ **bookkeeping** (the telescoping in §3 stands on this):

delay chain: $\delta h_1[k] = \delta h[k-2],\ \ \delta h_2[k] = \delta h[k-1],\ \ \delta h_3[k] = \delta h[k]$ (current, leading)

measurement: $y_1[k] = \delta x_m[k] = \delta h[k-d] + n_1$ observes δh_1, $d = 2$; $y_2[k] = a_{xm}[k]$ observes $\hat{a}_h$

per step: $\underbrace{\text{predict}}_{\delta\hat{h}_3[j+1]=\lambda_c\,\delta\hat{h}_3[j]} \longrightarrow \underbrace{\text{update}}_{\text{innovation } u}$

1. True (Track A)

a_h follows the real height. With $h[k{+}1]-h[k] = \Delta h_d + (1-\lambda_c)\delta h + F_{dh}e_{a_h} + \varepsilon_h$, a first-order expansion (drop a'') gives

$$a_h[k+1] \approx a_h[k] + a'_{hd}\,\Delta h_d + (1-\lambda_c)\,a'_{hd}\,\delta h + a'_{hd}\,F_{dh}\,e_{a_h} + a'_{hd}\,\varepsilon_h$$

$$\boxed{a'_{hd}[k+1] \approx a'_{hd}[k], \qquad a_r[k] = a_h[k] - a_h(h_d[k]) = -\,a'_{hd}[k]\,\delta h[k]}$$

The anchor $a_r = -a'_{hd}\,\delta h$ is the whole channel's footing: taking variances,

$$a'_{hdm}[k] := +\sqrt{\mathrm{Var}(a_r[k])\,/\,\mathrm{Var}(\delta h[k])} = a'_{hd}[k] \qquad (a'_{hd} > 0)$$

In the True track the ratio recovers a'_{hd} exactly. This is the gold standard the online version is judged against (verified: fig., left panel, a_r overlays $-a'_{hd}\,\delta h$).

2. Estimate (Track B)

Online there is no true $a_r, \delta h$ – only the EKF estimates $\hat{a}_h, \delta\hat{h}_3$. The same recipe is rebuilt from estimates with two EWMA stages each:

$$
\begin{aligned}
\overline{a}_h[k+1] &= (1-a_{var})\,\overline{a}_h[k] + a_{var}\,\hat{a}_h[k+1], \qquad \hat{a}_r[k] = \hat{a}_h[k] - \overline{a}_h[k] \\
\hat{\sigma}^2_{\hat{a}_r}[k+1] &= (1-a_{cov})\,\hat{\sigma}^2_{\hat{a}_r}[k] + a_{cov}\,\hat{a}_r^2[k+1] \\
\overline{\delta\hat{h}}[k+1] &= (1-a_{var})\,\overline{\delta\hat{h}}[k] + a_{var}\,\delta\hat{h}_3[k+1], \quad \delta\hat{h}_r[k] = \delta\hat{h}_3[k] - \overline{\delta\hat{h}}[k] \\
\hat{\sigma}^2_{\delta\hat{h}}[k+1] &= (1-a_{cov})\,\hat{\sigma}^2_{\delta\hat{h}}[k] + a_{cov}\,\delta\hat{h}_r^2[k+1]
\end{aligned}
$$

$$
\boxed{\hat{a}'_{hdm}[k] := +\sqrt{\hat{\sigma}^2_{\hat{a}_r}[k]\,/\,\hat{\sigma}^2_{\delta\hat{h}}[k]}}
$$

This is the number that would feed the filter. The predict of $\hat{a}_h$ (§4) uses $\hat{a}'_{hd}$ itself – the seed of the mirror.

3. Telescoping → mirror (predict-only bridge)

In a hold ($\Delta h_d = 0$), predict-only ($u \equiv 0$), the gain-state predict is

$$
\hat{a}_h[k+1] = \hat{a}_h[k] + (1-\lambda_c)\,\hat{a}'_{hd}\,\delta\hat{h}_3[k], \qquad (1-\lambda_c)\,\delta\hat{h}_3[j] = \delta\hat{h}_3[j] - \delta\hat{h}_3[j+1]
$$

using $\delta\hat{h}_3[j+1] = \lambda_c\,\delta\hat{h}_3[j]$ (§0 predict). Summing from k_0 telescopes the middle terms:

$$
\hat{a}_h[k] = \left[\hat{a}_h[k_0] + \hat{a}'_{hd}\,\delta\hat{h}_3[k_0]\right] - \hat{a}'_{hd}\,\delta\hat{h}_3[k] = C - \hat{a}'_{hd}\,\delta\hat{h}_3[k]
$$

The EWMA mean $\overline{a}_h$ removes the constant C, leaving

$$
\boxed{\hat{a}_r[k] = -\,\hat{a}'_{hd}[k]\,\delta\hat{h}_r[k] \qquad \Longrightarrow \qquad \hat{a}'_{hdm}[k] = \sqrt{\hat{a}'^{\,2}_{hd}\,\hat{\sigma}^2_{\delta\hat{h}}/\hat{\sigma}^2_{\delta\hat{h}}} = \hat{a}'_{hd}[k] \quad \forall\,\hat{a}'_{hd}}
$$

Mirror: the numerator $\hat{\sigma}^2_{\hat{a}_r}$ was built from $\hat{a}_h$, which the controller integrated using its own $\hat{a}'_{hd}$; so it reads $\hat{a}'^{\,2}_{hd}$ (your belief), not $a'^{\,2}_{hd}$ (truth). The shared $\delta\hat{h}$ cancels in the ratio and returns your input. No truth enters.

4. $u \neq 0$: the crack κ, and where y_1 contributes

With the real filter ($u \neq 0$) the innovation correction adds a term u_r (truth enters only through the y_2 innovation $\nu_2 \approx -e_{a'_h}\,\delta h_r[k-d] + n_2$):

$$
\hat{a}_r[k] = -\,\hat{a}'_{hd}\,\delta\hat{h}_r[k] + u_r[k]
$$

$$
\mathrm{E}[\hat{a}_r^2]/\sigma^2_{\delta h} = \hat{a}'^{\,2}_{hd} + \underbrace{2\hat{a}'_{hd}\,\kappa\,e_{a'_h}}_{\text{truth}} + \underbrace{\sigma_u^2/\sigma^2_{\delta h}}_{\text{floor}}, \qquad \mathrm{E}[\delta\hat{h}_r\,u_r] =: -\kappa\,e_{a'_h}\,\sigma^2_{\delta h}
$$

Linearising the square root, $\sqrt{\hat{a}'^{\,2}_{hd} + \epsilon} \approx \hat{a}'_{hd} + \epsilon/(2\hat{a}'_{hd})$,

$$
\boxed{\hat{a}'_{hdm}[k] = \hat{a}'_{hd}[k] + \kappa[k]\,e_{a'_h}[k] + n_{var}[k], \qquad n_{var} = b_u + w, \quad b_u = \frac{\sigma_u^2}{2\,\hat{a}'_{hd}\,\sigma^2_{\delta h}}}
$$

κ is the crack through which truth enters ($\kappa = 1$ measurement, $\kappa = 0$ pure mirror); $b_u > 0$ is the sqrt-rectification floor, $\propto 1/\hat{a}'_{hd}$.

Observability (y_1's role): y_1 is not absent – it *shapes* $\delta\hat{h}_r$, the signal shared by numerator and denominator, so it reinforces the mirror rather than breaking it. But in a hold y_1 is *blind* to a': $e_{a'_h}$ reaches δh only through $F_e(4,5) = \Delta h_d$ (§6), which is gated to 0; hence y_1 contributes nothing to κ. All of κ (tiny) comes from y_2.

5. Error (Track C = A − B)

Subtract the two a' updates. Truth: $a'_{hd}[k{+}1] = \beta a'_{hd} + (1-\beta)a'_{hdm}$ with $a'_{hdm} \equiv a'_{hd}$; estimate: $\hat{a}'_{hd}[k{+}1] = \beta\hat{a}'_{hd} + (1-\beta)\hat{a}'_{hdm}$. With $a'_{hdm} - \hat{a}'_{hdm} = (1-\kappa)e_{a'_h} - n_{var}$ from §4,

$$\boxed{\begin{aligned} &e_{a'_h}[k+1] = \big[\,1-(1-\beta)\,\kappa[k]\,\big]\,e_{a'_h}[k] - (1-\beta)\,n_{var}[k] \\ &e_{a_h}[k+1] = \big(1+\hat{a}'_{hd}F_{dh}\big)e_{a_h} + \Delta h_d\,e_{a'_h} + (1-\lambda_c)\hat{a}'_{hd}\,e_h + \hat{a}'_{hd}\,\varepsilon_h \end{aligned}}$$

Pole $= 1-(1-\beta)\kappa$. With $\kappa \approx 0$ the pole is 1: $e_{a'_h}$ is a pure *integrator* with no restoring force, pushed by the positive bias n_{var}. Open loop (shadow) $\hat{a}'_{hd}$ is fixed and $\hat{a}'_{hdm}$ just reads high; *closed loop* the integrator makes $\hat{a}'_{hd}$ ramp – bounded in a hold ($b_u \propto 1/\hat{a}'_{hd}$ self-limits, $\rightarrow 3\times$) but divergent under motion (taylor FF destabilises $\hat{a}_z$, $t \approx 0.8\,$s). The divergence is the derivation's own prediction, not a bug; hence **never close the loop**.

6. $\mathbf{F}_e$ (5×5)

$$x_e = [\,\delta h_1,\ \delta h_2,\ \delta h_3,\ e_{a_h},\ e_{a'_h}\,]^{\mathsf{T}}$$

$$\boxed{\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & (1-\lambda_c)\,\hat{a}'_{hd}[k] & 1+\hat{a}'_{hd}[k]F_{dh}[k] & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & 1-(1-\beta)\,\kappa[k] \end{bmatrix}, \quad q = \begin{bmatrix} 0 \\ 0 \\ -\varepsilon_h \\ \hat{a}'_{hd}\,\varepsilon_h \\ -(1-\beta)n_{var} \end{bmatrix}}$$

Row 4 col 5 $= \Delta h_d$ is the only path $e_{a'_h} \rightarrow e_{a_h}$: open only under motion. Row 5 diagonal is the §5 pole.

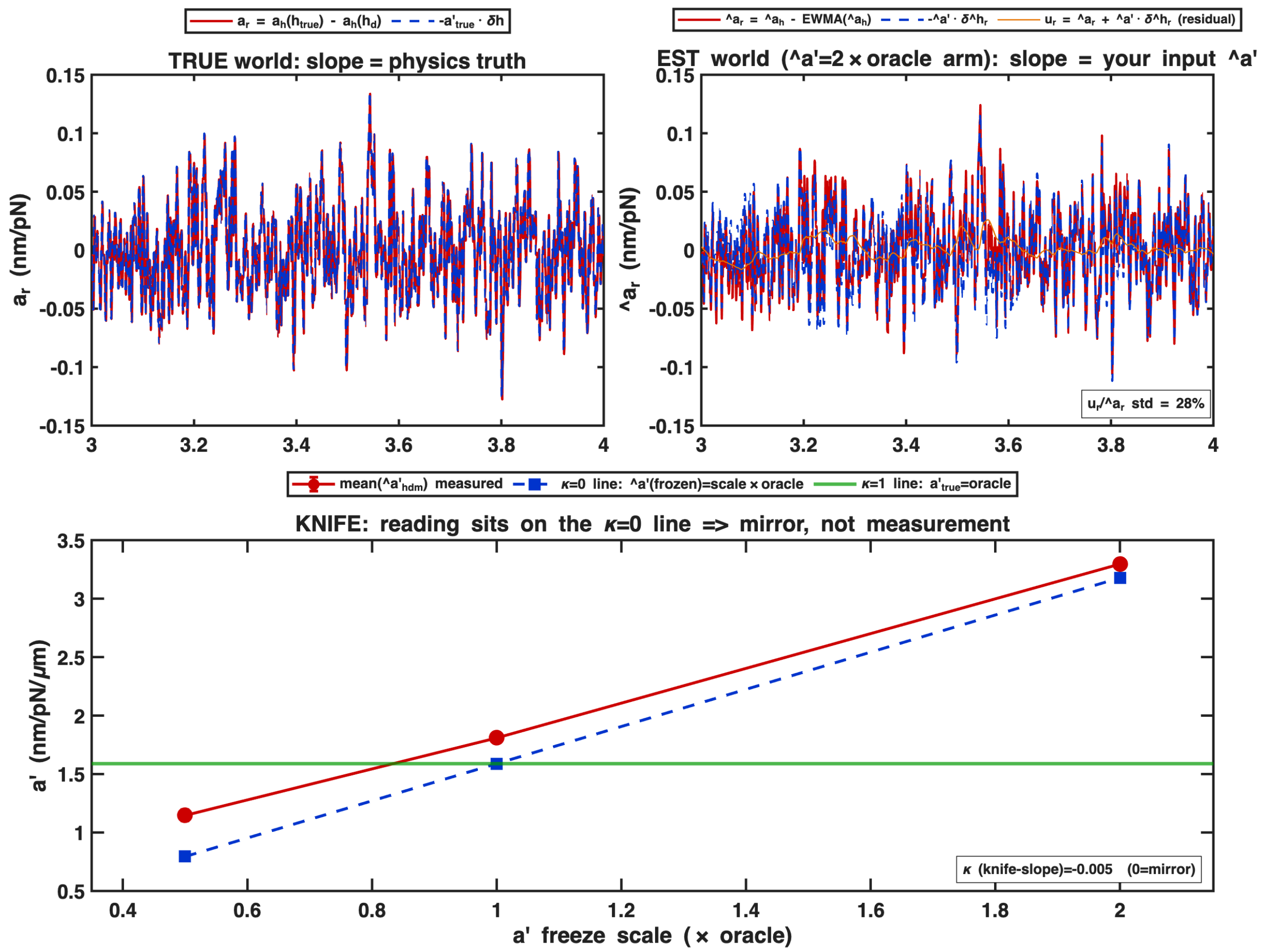
7. Verification (frozen-arm knife)

Setup: pure HOLD at $\bar{h} = 2.0$, *full* EKF (taylor-gain 4-state, $y_2 = a_{xm}$ plain, $u \neq 0$ throughout). The slope $\hat{a}'_{hd}$ is frozen at $\{0.5, 1, 2\}\times$ the oracle a'_{hd} (`aprime_source='known'`+`aprime_scale`) – the as-state architecture with a' pinned. (Live as-state cannot be used: in a hold $\Delta h_d = 0$ makes a' unobservable, so slot-5 stays ≈ 0 and there is no mirror term.) The reading $\hat{a}'_{hdm}$ is only computed, never fed back.

Discriminator: ${\hat{a}'_{hdm}}^2 - {\hat{a}'_{hd}}^2 = 2\hat{a}'_{hd}\kappa(a'_{hd} - \hat{a}'_{hd}) + \sigma_u^2/\sigma_{\delta h}^2$. $\kappa = 0$ gives a *constant* floor across arms; $\kappa = 1$ a sign-flipping term. Measured (4 seeds):

$\hat{a}'_{hd}$ scale	mean $\hat{a}'_{hdm}$ [nm/pN/μm]	${\hat{a}'_{hdm}}^2 - {\hat{a}'_{hd}}^2$	(κ=1 would give)
0.5×	1.15	$+6.83 \times 10^{-7}$	$+1.26 \times 10^{-6}$
1.0×	1.81	$+7.48 \times 10^{-7}$	0
2.0×	3.30	$+7.62 \times 10^{-7}$	-1.01×10^{-5}

The floor is constant ($\approx 7 \times 10^{-7}$), not sign-flipping: knife-slope fit $\kappa = -0.005 \approx 0$ (mirror, floor-immune). The reading tracks the frozen $\hat{a}'_{hd}$, not the truth. The relative inflation (+44% at 0.5×, +3.7% at 2×) is the $b_u \propto 1/\hat{a}'_{hd}$ floor (§4). The mirror tightens with $\hat{a}'_{hd}$ (residual $u_r/\hat{a}_r$: 76% → 51% → 28%).



Left (TRUE world): $a_r = a_h(h_{true}) - a_h(h_d)$ overlays $-a'_{hd}\,\delta h$ – slope is physics truth. Right (EST world, $\hat{a}'_{hd} = 2\times$oracle arm): $\hat{a}_r$ overlays $-\hat{a}'_{hd}\,\delta\hat{h}_r$ (residual u_r in orange) – slope is your input $\hat{a}'_{hd}$, not truth. Bottom (KNIFE): the reading (red) sits on the $\kappa = 0$ line (frozen $\hat{a}'_{hd}$), off the $\kappa = 1$ line (oracle truth); $\kappa = -0.005$.